DSCI441 COURSE POSTER

Used Car Price Prediction System

End-to-end Craigslist price regression pipeline with baseline comparison, nonlinear modeling, extension diagnostics, and deployment-facing web demo.

BEST BROAD-DOMAIN R²
0.8061
HistGradientBoosting (`hgb_ordinal`)

BEST BROAD-DOMAIN RMSE
$6,649
Test split (80/20, seed 441)

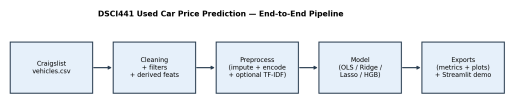
EXTENSION DIAGNOSTICS
6 figures
Subgroup + decile diagnostics

PROBLEM DESCRIPTION / MOTIVATION

Why used-car price regression is not a single-metric task

Price prediction must handle heavy tails, missingness, and nonlinear segment interactions. A single global RMSE can hide large differences across price ranges and vehicle segments.

- Target is strongly right-skewed.
- Many categorical fields are sparse/missing.
- Error scale increases with true price (heteroskedasticity).



Consistent workflow from cleaning to model export and interactive demo.

DATASET DESCRIPTION

Craigslist listings with mixed tabular structure

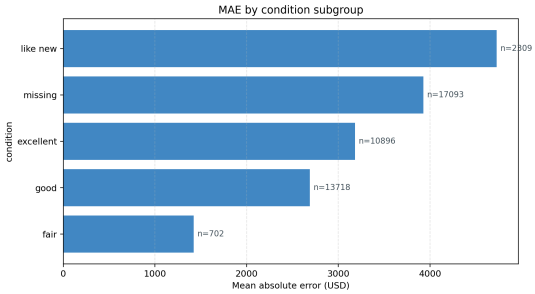
The dataset contains 426,880 raw listings with numeric, categorical, and optional text fields. Cleaning and feature derivation produce a stable training table for fair model comparison.

426,880
Raw listings

17+
Core input attributes

Heavy-tail
Price distribution shape

Missingness
High in several categorical columns



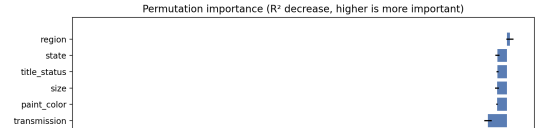
Extension diagnostics confirm subgroup-level MAE variation.

EXPERIMENTS / MODEL SELECTION

Linear baselines, text-augmented ridge, and boosted trees

Experiments include OLS, Ridge, Lasso, Ridge+TF-IDF, and HistGradientBoosting. The nonlinear model gives the strongest broad-domain fit while preserving a reproducible pipeline.

Boosted trees lead in broad-domain R².



EVALUATION

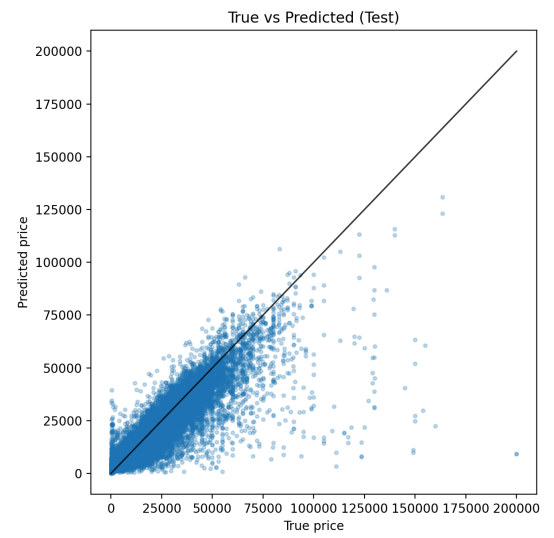
Global fit is strong, but error behavior is segment-dependent

The best model achieves high broad-domain fit, yet error distribution differs by price decile and subgroup. Extension diagnostics expose these deployment-relevant patterns.

- The headline model is strong enough to summarize the broad market, but not uniform enough to justify one confidence message.
- High-price listings remain the most visible source of larger absolute-dollar miss.
- The extension therefore shifts the project from score reporting to decision-aware evaluation.

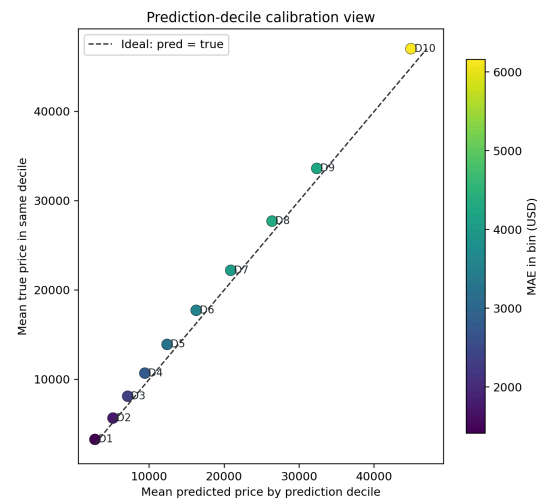
Broad-domain best: R² 0.8061, RMSE 6649, MAE 3384.

Decile drift: MAE rises from about $1.8k to about $10.1k (D1 -> D10).



High-price tail remains harder even with best overall model.

MAE and mean residual shift by true-price decile.



Decile-level predicted vs true means reveal calibration-style drift.